From High-Resolution Macroburst Observations to WRF-Based Mechanism Reconstruction: A Critical Review of Mesoscale Modeling for Destructive Convective Winds

Project-local candidate for PI review. Scientific status: `not_approved`.

Abstract

Destructive convective winds occupy a difficult interval between storm-scale prediction and engineering-scale description. Mesoscale models can reconstruct the environment, convective organization, and candidate acceleration processes, while the most consequential surface winds may vary over seconds, hundreds of meters, and instrument-dependent sampling volumes. This study uses Childs et al. (2021) as a recurring observational anchor for evaluating Weather Research and Forecasting (WRF) model studies of macrobursts, downbursts, gust fronts, and related convective wind systems. The Akron, Colorado, macroburst of 9 June 2020 supplies a demanding validation vector built from instrument-specific wind, pressure, temperature, direction, radar morphology, and damage evidence. Schumacher et al. (2023) provide the bounded mechanism bridge from that vector to a real-case WRF reconstruction and a separate idealized CM1 experiment, in which gravity-wave trapping, amplification, and breaking constitute a physically plausible, event-bounded candidate mechanism. Across the 28-source review corpus, WRF commonly represents a parent convective system, broad timing, or environmental structure more credibly than a local extreme gust, its pressure-temperature sequence, or its exact footprint. Finer nominal grid spacing can expose relevant structure without guaranteeing usable effective resolution, correct triggering, or realistic small-scale energy. A defensible WRF macroburst study consequently needs observation operators for every comparator, separate tests of event identity, morphology, local wind, thermodynamic sequence, and mechanism, and explicit uncertainty across scales. The resulting evidence chain treats modeling as a sequence of partial

matches, with engineering admissibility evaluated separately from meteorological reconstruction. Scientific status is `not_approved`, and PI review is required before any configuration, mechanism, transfer, or engineering interpretation is promoted.

1 Introduction

Destructive convective winds are often grouped by consequence even when their spatial support, duration, parent storm, and physical acceleration differ. The observational definitions introduced by Fujita (1985) distinguish microbursts from macrobursts by damaging-outflow scale, while the broader derecho literature identifies long, organized convective wind swaths rather than a single local downdraft. Johns and Hirt (1987) used sustained swath and convective-organization criteria to characterize derechos, and Ashley and Mote (2005) treated derecho occurrence and consequences at national and event scales. Neither framework establishes the local process that produced a maximum wind at a particular point. WRF evaluation becomes ambiguous when a broad parent-system match is presented as local gust fidelity.

WRF is valuable because it connects synoptic forcing, mesoscale boundaries, storm initiation, microphysical tendencies, near-surface flow, and three-dimensional diagnostics within a common numerical system. The ARW technical architecture described by Skamarock et al. (2019) supports nonhydrostatic dynamics, real-data initialization, nested domains, multiple physics interfaces, and data assimilation. That flexibility creates analytical opportunity and a large configuration space. A successful reconstruction reflects interacting choices of forcing data, initialization time, lateral-boundary treatment, grid design, vertical discretization, microphysics, planetary boundary layer treatment, land surface, assimilation, filtering, and output cadence. Technical availability of an option offers no event-specific validation. Reproducibility consequently requires more than a model name and innermost grid spacing.

The validation problem is most severe near the ground. A numerical grid cell is a filtered volume, a reported 10 m wind may be diagnostic, a lowest-model-level wind belongs to another height, and a model snapshot is not an instrument gust. Field observations also have distinct supports. A high-rate sonic anemometer, a conventional ASOS report, a radar radial velocity, a profiler retrieval, and a damage vector integrate different spatial, temporal, and physical information. Hjelmfelt (1988) showed that observed microburst outflows are asymmetric and evolve rapidly, while Wakimoto (1982) separated hydrostatic cold-pool pressure from a prefrontal nonhydrostatic contribution across gust-front life-cycle stages. He et al. (2024) further showed that profiler and near-ground records can expose vertically varying and nonstationary wind structure that a single surface peak cannot represent. These observations define comparison targets with their own measurement limits.

The 9 June 2020 Akron record makes the scale mismatch unusually explicit. Childs et al. (2021, p. 1) reported 51.12 m s^{-1} at a 2.3 m eddy-covariance tower and 50.45 m s^{-1} at an adjacent 10 m tower. The same page records a 5 hPa pressure surge in 19 s immediately after the strongest winds, followed by a 15 hPa decrease during the next 3 min. Temperature rose by 1.5 °C in less than 1 min concurrent with the maximum gusts, then fell by 3.5 °C during the following minute. These measurements replace vague station counts and turn the event into an ordered multivariable trajectory. Timing, phasing, instrument height, sampling support, and footprint become part of the target rather than annotations to one maximum wind.

Schumacher et al. (2023) converted that observational challenge into a numerical mechanism study. Its real-case WRF reconstruction and separate idealized CM1 experiment supported a trapped gravity-wave amplification and breaking interpretation for Akron. The interpretation is physically consequential and bounded by one event, specified environmental and numerical configurations, an imperfect station-level pressure comparison, and two models with different purposes. The wider review corpus tests that chain from several directions: direct WRF cases examine destructive winds in Shanghai, Zhejiang, Genoa, the Mediterranean, and MIST microbursts; resolution studies challenge the assumption that nominal refinement equals usable information; and TASS, ARPS, idealized cloud-model, observational, engineering-experiment, climatological, and machine-learning sources provide benchmarks, mechanisms, event context, or transfer warnings. This study organizes those sources around four distinct assessments: whether WRF reconstructs the parent event, whether it reconstructs the local destructive wind, whether its diagnostics support a mechanism, and whether its output can enter a smaller-scale engineering workflow. Figure 1 previews this evidence chain and shows the review gates that keep observation, real-case WRF, idealized CM1, wider comparison, and engineering transfer separate.

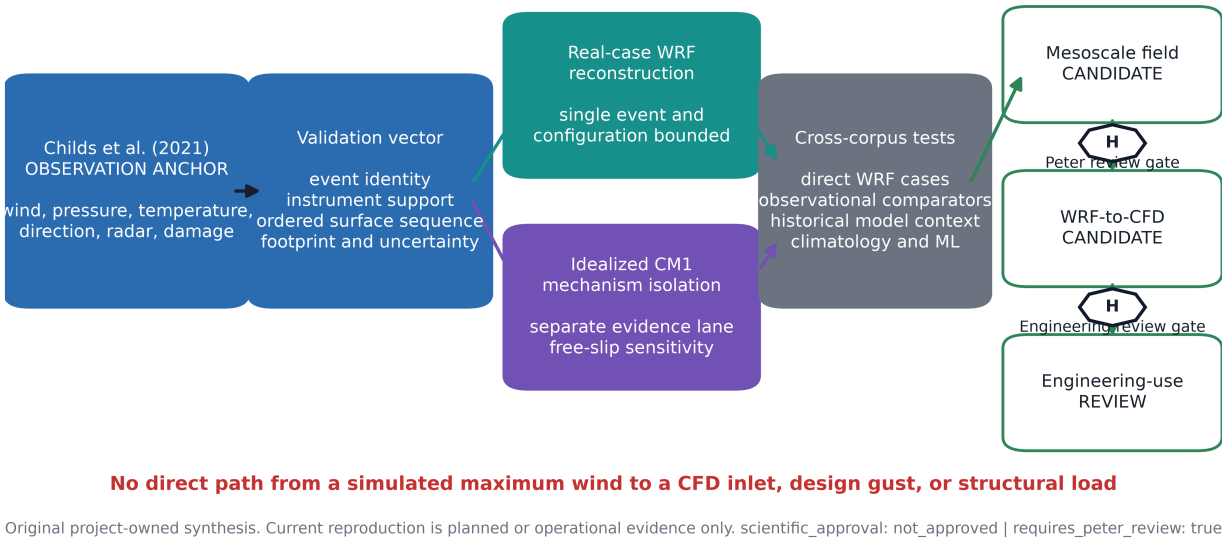


Figure 1: Childs-centered evidence chain. Childs et al. (2021) observations define the event and validation vector; Schumacher et al. (2023) separate real-case WRF reconstruction from idealized CM1 mechanism isolation; the review corpus tests reconstruction, resolution, alternative mechanisms, and transfer; explicit scientific and PI gates precede any WRF-to-CFD candidate. Schematic traces are not digitized source data. Scientific status: `not_approved`.

2 Childs et al. (2021) as the observational anchor

The Akron record begins with event identity. Childs et al. (2021, p. 1) documented a destructive macroburst near Akron, Colorado, on 9 June 2020. The event occurred with elevated convection above a stable near-surface layer (Childs et al., 2021, pp. 4 to 7), complicating a default interpretation based on an ordinary surface-based storm and cold pool. USDA high-frequency stations, ASOS, radar, environmental analyses, forecast context, and damage evidence constrain different portions of the event. Station records directly constrain local time series at specified heights and sampling rates. Radar constrains parent-storm morphology and motion through beam-dependent observations. Damage provides integrated evidence of consequence and direction, while model-derived proximity soundings provide environmental context. Neither damage nor modeled context is a direct three-dimensional observation.

Wind measurements demonstrate why one peak cannot serve as the complete validation

target. Schumacher et al. (2023, p. 1), summarizing the Childs record, gives 37.04 m s^{-1} at ASOS, 50.45 m s^{-1} at the 10 m tower, and 51.12 m s^{-1} at the 2.3 m eddy-covariance tower sampled at 10 Hz. The Childs source record confirms the two tower maxima and identifies them by instrument (Childs et al., 2021, pp. 1, 16). Each value has a distinct height, response, and temporal representation. Comparing a WRF 10 m field with either tower requires a declared gust definition and averaging interval, while a lowest-level WRF wind or grid-cell maximum introduces different vertical and spatial filtering. Instrument-specific comparisons retain information lost in a generic threshold-based station count.

Pressure and temperature make the target more discriminating. The Childs abstract records a 5 hPa pressure rise in 19 s, a subsequent 15 hPa decrease over 3 min, a 1.5 °C rise in less than 1 min concurrent with the maximum gusts, and a 3.5 °C fall during the following minute (Childs et al., 2021, p. 1). The detailed trace shows that temperature began to rise first, the maximum gust followed, and the rapid pressure surge followed within the same minute (Childs et al., 2021, p. 13). A simulation that produces a broad pressure ribbon with approximately correct amplitude but incorrect order is a partial match. The same is true of a temperature perturbation with displaced phasing. Wind, pressure tendency, temperature, precipitation, and direction need a shared clock because their sequence distinguishes competing interpretations more sharply than independent extrema.

Direction and footprint add spatial tests. Childs et al. (2021, p. 1) described shifting wind direction together with the National Weather Service damage survey as suggestive of both radial outflow and a gust-front passage. This qualified inference is stronger than either evidence type alone but is not an instantaneous velocity map. A simulated maximum near Akron is incomplete if its footprint is displaced, its directional sequence conflicts with the observations, or its parent system reaches the site at the wrong time. Direction needs a common coordinate frame, and storm-motion-relative quantities must remain distinct from earth-relative records.

Radar morphology supplies another component without resolving mechanism. The Childs record constrains elevated convection, banding, system motion, and the relationship between the parent storm and local response. Borchardt et al. (2024, pp. 7 to 11) reinforce this distinction across elevated-convection episodes: reflectivity intensity and spacing alone did not separate damaging from nondamaging cases, whereas combined radar-velocity and surface sequences were more informative.

The environmental component includes stable near-surface stratification, elevated convective heating, low-level shear, an inversion, and possible critical-layer structure. RAP and WRF proximity profiles are modeled context, not realized soundings at the event time. Agreement between modeled profiles does not create independent environmental validation. When

a WRF profile seeds CM1, it becomes an input to a mechanism experiment and not a second observation.

The resulting validation vector has linked but separable dimensions: event time and position, parent-storm morphology and motion, wind footprint, instrument-specific gust magnitude and duration, pressure trajectory, temperature trajectory, directional evolution, environmental stratification, vertical structure, and candidate mechanism diagnostics. Passing some dimensions and failing others locates weak evidence links more effectively than a binary success label. Figure 2 places these dimensions on synchronized observation and model-comparison lanes so that height, cadence, spatial support, and unresolved quantities remain visible.

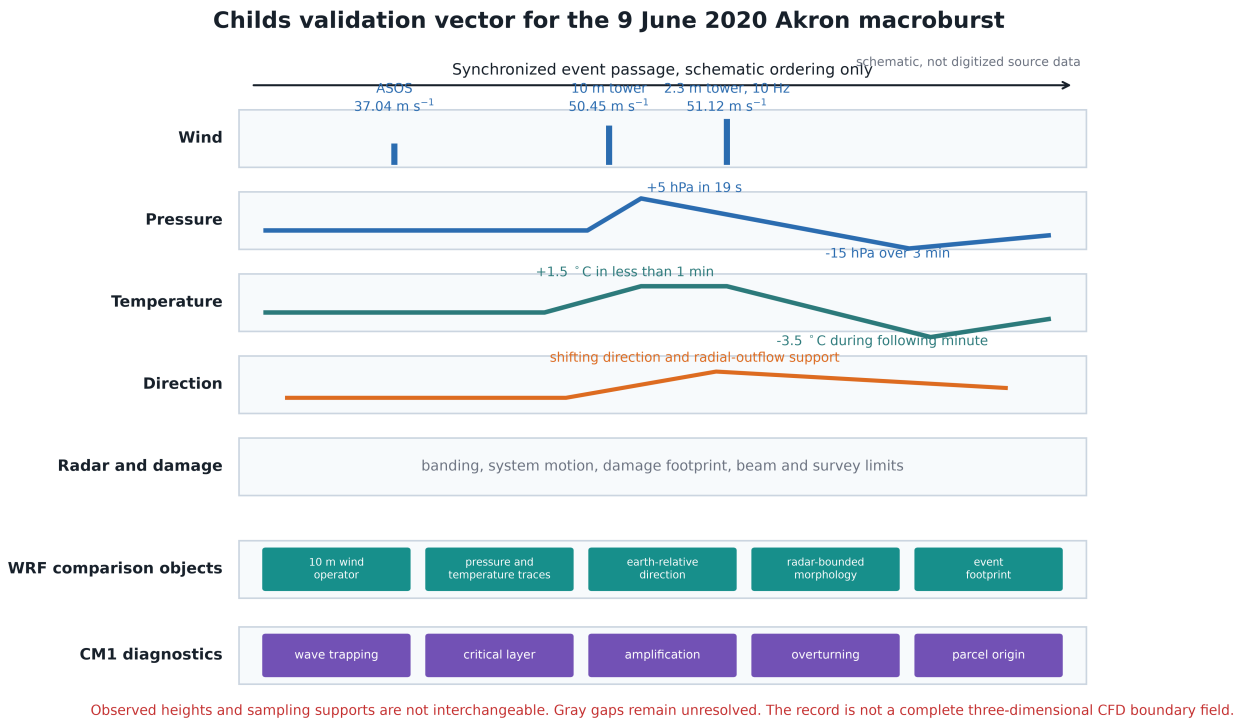


Figure 2: Synchronized Childs validation vector. Instrument-specific wind records (37.04, 50.45, and 51.12 m s^{-1} at ASOS, 10 m tower, and 2.3 m 10 Hz tower), the corrected pressure sequence (5 hPa rise in 19 s, 15 hPa decrease over 3 min), and the temperature sequence (1.5 °C rise in less than 1 min, 3.5 °C fall during the following minute) are placed on a common timeline. Adjacent lanes identify admissible WRF comparison variables and separate CM1 diagnostics. Gray regions mark dimensions the observations do not resolve. Schematic traces are not digitized source data.

The observation anchor also defines what is missing. Point stations, radar, damage, and modeled context do not constitute a complete three-dimensional inflow field. Turbulence between instruments, spatially unobserved regions, pressure continuity over a boundary plane,

and engineering-scale coherence are unresolved. Observational richness raises the standard for WRF evaluation but does not eliminate the need for observation operators and uncertainty analysis. The record in Childs et al. (2021) anchors auditable comparisons but supplies neither direct numerical forcing nor a completed engineering field.

3 Schumacher et al. (2023): the WRF/CM1 mechanism bridge

Schumacher et al. (2023, p. 4) start from the Akron constraints and ask how elevated convection in a stable, strongly sheared postfrontal environment could produce surface winds stronger than those available in the ambient vertical profile. WRF-ARW version 4.0.3 carries real-event forcing, nesting, physics parameterizations, terrain, and evolving mesoscale context. CM1 uses a controlled, horizontally homogeneous environment to isolate dynamics. The models address one scientific question through different evidence identities, so agreement can strengthen plausibility without allowing either model to validate the other.

The WRF configuration makes the scale transition explicit. The outer domain used 4 km grid spacing, 51 vertical levels, and a 25 s time step, while the innermost nest used 267 m spacing, 80 levels, and a 1.33 s time step (Schumacher et al., 2023, p. 4). The simulation used MYJ PBL treatment, Thompson microphysics, and GFS forecast forcing. The outer domain missed the severity of the surface winds. The 267 m nest produced swaths above 30 m s^{-1} and localized downbursts above 45 m s^{-1} (Schumacher et al., 2023, p. 7). This event-specific contrast supports the need to expose finer structure without identifying 267 m as a universal threshold or equating nominal spacing with effective resolution.

The WRF diagnostics connect storm heating to wave dynamics. Elevated latent heating generated a gravity-wave response in an environment with a stable layer below and a critical layer aloft. Wave energy became trapped, amplitude increased, and nonlinear breaking generated strong downward acceleration. Trajectories feeding the intense downdrafts originated mainly within the lowest 1 km in the reported reconstruction (Schumacher et al., 2023, p. 9). A rapid increase in the Scorer parameter just above 4 km AGL identified the critical-layer region that restricted upward energy propagation (Schumacher et al., 2023, p. 10). These diagnostics test wave trapping, parcel origin, and acceleration beyond a morphology comparison, while remaining dependent on the simulated environmental profile and numerical representation.

Comparison with the Childs vector is mixed. High-resolution WRF generated wave-like convection, strong near-surface winds, and low-high-low pressure organization consistent with

portions of the observed sequence. It also produced a pressure decrease before the extreme rise, whereas the station record showed nearly constant pressure before that rise (Schumacher et al., 2023, p. 8, Fig. 8). This difference changes the ordered validation target and is not a minor timing offset. A broad pressure structure can be compatible with wave activity while local phasing is wrong. Simulated severe wind near Akron likewise differs from reproducing each instrument-specific maximum, duration, height, and footprint.

CM1 provides the mechanism-isolation branch. Its 125 m horizontal spacing, 110 stretched vertical levels, 0.5 s time step, and 1200 by 1200 horizontal grid supported an idealized warm-bubble experiment under horizontally homogeneous initial conditions (Schumacher et al., 2023, pp. 4, 6). Trapped-wave amplification and breaking produced repeated surface impacts. Peak gusts exceeded 70 m s^{-1}, substantially above the Akron observations, and the free-slip lower boundary was identified as an important cause (Schumacher et al., 2023, p. 10). The experiment strengthens the dynamic chain while exposing why idealized intensity cannot be transferred to the observed event.

The bridge is most informative when every link stays explicit. Childs et al. (2021) supply event identity and the synchronized multivariable sequence. Real-case WRF tests whether a specified reconstruction can produce the parent system, environmental structure, and candidate process near the event. CM1 tests whether the proposed dynamics can arise under simplified conditions. Interpretation then compares agreement, conflict, and missing quantities. Engineering use remains outside both studies because neither supplies an independently validated boundary plane, a design gust, or a structural load.

A broader elevated-convection sample bounds how far the Akron interpretation might extend. Borchardt et al. (2024, pp. 4 to 11) cataloged 52 episodes under study-specific operational definitions, with six representative damaging cases showing combinations of leading-edge velocity divergence, pressure oscillations, motion-aligned wind changes, and steady or warmer temperatures. Its 12 May 2022 WRF case used 3 km and 600 m nests and qualitatively reconstructed banding and wave-like surface response, but the system evolved too slowly, widespread severe wind was underrepresented, and clear wave breaking was absent (Borchardt et al., 2024, pp. 11 to 17). The comparison supports a broader family of wave-coincident events while preserving breaking as an event-conditioned hypothesis.

The Schumacher bridge supplies a testable explanation for observations that challenge a minimal cold-pool narrative and a method for pairing real-case and idealized models while preserving their distinct roles. Current PETW reproduction activity documents planning and workflow readiness, not scientific evidence. Scientific validation requires a source-locked run, independently checked outputs, and an event-level comparison against the Childs vector. The next issue is whether the Akron distinctions persist across other WRF reconstruction

regimes.

4 WRF representations of destructive convective winds beyond Akron

Cross-event comparison first separates parent-system identity from local convective initiation. Burlando et al. (2024, pp. 6 to 8) provide the clearest paired outcome: a WRF version 4.1.1 hindcast with a 2.5 km finest grid and radar-lightning assimilation captured the broader 18 August 2022 derecho landfall and general evolution while failing to initiate a smaller damaging Ligurian cell several hours earlier. The represented regional system and the missed local storm belong to one episode but are different validation targets. The reconstruction represented portions of the derecho morphology and regional timing but did not initiate the earlier local event. For Akron, this outcome shows why a favorable parent-system match cannot substitute for the destructive wind at the target site.

The Genoa case reaches a similar limit through a different experimental design. Burlando et al. (2020, pp. 6 to 7) executed WRF v3.8.1 without assimilation on 13.5, 4.5, and 1.5 km nests using GFS or IFS forcing. The assimilating WRFDA v3.9.1 branches instead used 22.5, 7.5, and 2.5 km nests with 3DVAR radar-reflectivity assimilation. Five-minute maximum 10 m wind was compared at eight port anemometer locations and not at the bridge (Burlando et al., 2020, p. 23). WRF-IFS-DA produced near-surface winds near 20 m s^{-1} from 11:00 to 11:55 UTC (Burlando et al., 2020, p. 24), but simulated storm development was delayed across the tested branches (Burlando et al., 2020, pp. 28 to 29). The assimilating IFS branch had the smallest delay, while the first three configurations underestimated maximum velocity during the bridge-collapse hour. Selected local wind metrics improved in the IFS-DA branch, but storm timing was delayed and the observations did not sample the bridge exposure. These limitations leave local meteorological reconstruction and bridge causality unresolved.

Local gust magnitude and footprint form a second comparison axis. The Shanghai reconstruction used WRF v4.1.5 with two one-way `ndown` domains at 1 km and 333.3333 m, 60 vertical layers, hourly ERA5 initial fields, Morrison two-moment microphysics, and YSU PBL treatment (Liu et al., 2023, p. 3). NCEP UPP v4.1 supplied the gust diagnostic (Liu et al., 2023, p. 5). Simulated strong-wind centers were concentrated at 20:06 and 20:18 BLT, while observed extremes generally occurred earlier; modeled speeds were especially high during decay (Liu et al., 2023, p. 6). The simulated echo and wind support was broader, with direction differences and persistent decay-stage excess (Liu et al., 2023, p. 7). A maximum simulated horizontal wind of 23 m s^{-1} occurred at 20:06 BLT (Liu et al., 2023, p. 8). The Shanghai

and Zhejiang results together show that process diagnostics can be informative even when footprint or intensity fails a local comparator, provided that the mismatch remains part of the claim.

The Zhejiang case presents the complementary error of useful structure with underestimated intensity. Li et al. (2023, pp. 3 to 4) used WRF v4.4 with one-way nests at 3 km, 1 km, and 333.3 m, 51 terrain-following levels, ERA5 forcing, WSM6 microphysics, and YSU PBL treatment. Major mesoscale and convective structures were reconstructed, whereas maximum surface wind remained too weak. Trajectories and a kinetic-energy budget implicated evaporative cooling, pressure-gradient work, vertical energy transport, and outward horizontal advection. Akron differs from both East Asian cases in environmental and thermodynamic signature: its elevated convection, stable near-surface air, short warming followed by cooling, and rapid pressure sequence constrain a wave-related interpretation that the Shanghai and Zhejiang analyses neither refute nor generalize.

Physics sensitivity provides a third axis but does not yield a transferable package ranking. Bolgiani et al. (2020a, pp. 14, 20 to 21) tested three MIST microburst days with a 400 m innermost WRF grid, 3 min output, and a matrix of Morrison versus WSM6 microphysics and MYJ versus MYNN PBL treatment. Morrison branches achieved better source-reported event-count and detection agreement than corresponding WSM6 branches, while different Morrison-PBL pairings favored downburst-variable and outflow-variable counts. Simulated downdrafts and gusts were too weak, with peak simulated wind near 12 m s^{-1} and no gust exceeding 25 m s^{-1} (Bolgiani et al., 2020a, p. 20). Minimum simulated vertical velocity generally exceeded -4 m s^{-1}, indicating weaker downdrafts than the characteristic value near -12 m s^{-1} (Bolgiani et al., 2020a, p. 20). Microphysics and PBL treatment affect several Childs-vector components simultaneously: microphysics changes latent heating and cooling, hydrometeor loading, precipitation timing, pressure, and buoyancy, while PBL treatment changes near-surface mixing, shear, stratification, and drag. Cohen et al. (2015, pp. 18 to 19) found material PBL sensitivity in two southeastern United States cold-season severe-weather environments, although the 4 km design used proxy soundings and all tested schemes overestimated mixed-layer CAPE.

The same constraint appears in nonstandard process representation. Lompar et al. (2018, pp. 3 to 14) imposed periodic vertical forcing in the lowest model levels ahead of precipitation to represent gust-front head collapse. The modification changed cloud and precipitation evolution in an ideal supercell and one UAE event, yet both modified and default real-case simulations differed from observed radar location, geometry, and orientation. This result demonstrates feedback between imposed low-level forcing and convection within the tested setup, not a general repair for event initiation or destructive-wind fidelity.

Forcing and assimilation form the final comparison axis. GFS, IFS, ERA5, and HRRR appear across the direct cases, sometimes with radar or lightning assimilation. The Genoa IFS-DA branch improved selected local wind and delay behavior, whereas assimilation did not recover the earlier Ligurian cell in Burlando et al. (2024). GFS forecast forcing supported the configuration-bounded Schumacher reconstruction. Climatological verification reinforces why the comparison level must be explicit: Trapp et al. (2010, pp. 8 to 9) found that reanalysis-driven 4.25 km WRF integrations represented portions of regional and seasonal hazardous-convection occurrence through a combined updraft-helicity and reflectivity proxy, but strict climatological similarity emerged only after aggregation to several-hundred-kilometer scales. Cross-case synthesis indicates a consistent ordering of difficulty: environment and parent-system evolution are usually the most defensible targets; local initiation, path, and timing are less stable; instrument-height gust, duration, pressure phasing, and footprint are more demanding; and mechanism diagnosis requires vertical and budget evidence.

5 Resolution, effective resolution, and the convection gray zone

Grid spacing is one of the most visible WRF configuration values and one of the easiest to overinterpret. Schumacher et al. (2023, pp. 4, 7) showed a major difference between its 4 km outer domain and 267 m nest for Akron severity, while Bolgiani et al. (2020a, pp. 14, 20 to 21) used 400 m WRF for MIST microbursts and still underestimated downdraft intensity. Burlando et al. (2024, pp. 6 to 8) captured a large derecho but missed a small local cell at 2.5 km. Liu et al. (2023, pp. 3, 5, 6 to 8) and Li et al. (2023, pp. 3 to 17) obtained useful storm diagnostics at 333.3 m while retaining footprint or intensity error. These outcomes establish that some local structures require refinement, but they do not define a monotonic relation between nominal spacing and credibility.

Bolgiani et al. (2020b) directly separate nominal grid spacing, minimum sampling resolution, and effective resolution. The WRF version 4.0.3 experiment used nested spacings of 10.8 km, 3.6 km, 1.2 km, and 400 m, with correspondingly different output intervals (Bolgiani et al., 2020b, p. 2). Kinetic-energy spectra showed overenergetic fine scales in the 1.2 km and 400 m domains for this configuration and case set. The 3.6 km domain had a source-reported minimum resolution of 7.2 km and an effective resolution near 30 km under the study's diagnostic (Bolgiani et al., 2020b, pp. 8, 11). The 30 km result is not a universal WRF constant and not a physical microburst diameter. It demonstrates that grid spacing alone does not identify the smallest trustworthy atmospheric structure.

The convection gray zone compounds the problem. At intermediate spacing, energetic motions may be partly resolved while PBL and convective parameterizations still represent related subgrid processes. Turning a parameterization on or off does not guarantee consistency. The numerical discretization, diffusion, filters, time step, vertical grid, surface exchange, and physics coupling all shape the spectrum. Skamarock et al. (2019) describe these components and their interactions, including split-explicit time integration, grid staggering, mixing, filtering, nesting, and physics tendencies. The technical note defines what must be reported but does not certify a resolution for destructive winds.

Historical numerical studies offer useful physical benchmarks while underscoring model identity. Proctor (1988) used axisymmetric TASS, rather than WRF, and found that very fine grids were needed to represent the near-ground radial-velocity profile and burst-front structure of isolated microbursts. Proctor (1989) extended TASS sensitivity to precipitation loading, shaft size, hydrometeor type, duration, stable layers, and rotation under imposed forcing instead of a parent-cloud simulation. These studies motivate resolution and mechanism tests while supplying neither WRF validation nor a transferable WRF grid. Qu and Ji (2013) provide another cross-model analogue: its executed mesoscale model was ARPS, followed by a separate cloud model and FLUENT. The workflow demonstrates that refinement across models can alter intensity and structure, and that an environmental profile can improve a downstream cloud simulation even when the mesoscale model does not directly resolve convection at the event location.

Observation scale must be considered alongside effective model resolution. The Childs record includes changes over seconds, point measurements at different heights, radar volumes, and damage footprints. A WRF output interval of several minutes may miss the rate and ordering of the station trace even if the model internally produces a relevant disturbance. Saving output more frequently does not create spatial resolution or correct dynamics, but insufficient cadence can make validation impossible. Similarly, a 267 m cell does not correspond to a 2.3 m instrument height or a point sample. An observation operator must filter or aggregate both model and measurement into a common quantity. Figure 3 makes the scale ledger explicit by separating nominal spacing, diagnosed effective resolution, archive cadence, observational support, and supported claim scope rather than ordering studies by apparent fineness.

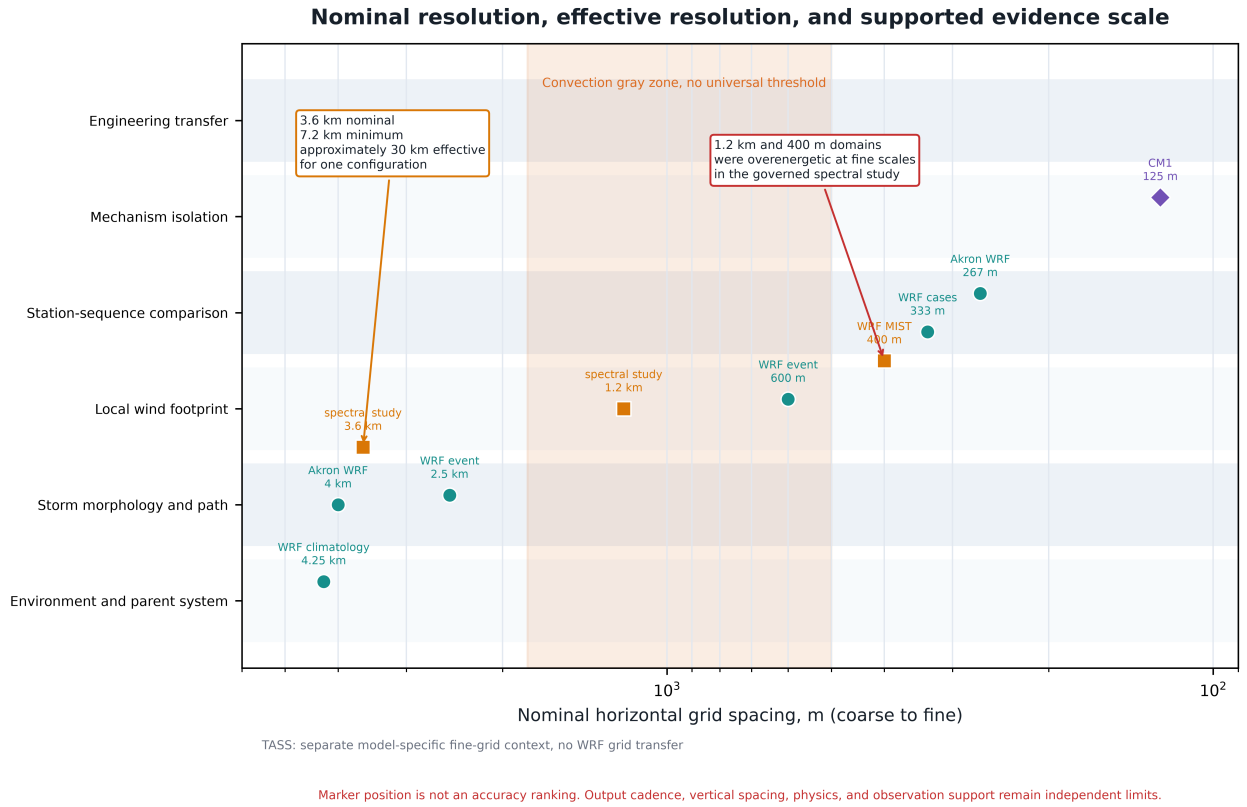


Figure 3: Nominal resolution versus supported evidence scale. Nominal spacings from kilometer-scale WRF through subkilometer WRF, 125 m CM1, and finer historical TASS cases are positioned by the evidence they can support. Separate bands show nominal spacing, diagnosed effective resolution (e.g., 3.6 km nominal yielding approximately 30 km effective for one configuration; Bolgiani et al., 2020b, pp. 8, 11), output cadence, observation support, and mechanism or engineering claims. The graphic avoids a monotonic accuracy arrow and marks the convection gray zone. Model identities (WRF, CM1, TASS) are not interchangeable.

The practical implication is not that fine grids are undesirable. The Akron reconstruction indicates that coarse grids can miss event severity, and several direct cases recover more local structure after refinement. The implication is that a refinement claim must be paired with spectral or convergence evidence, physics-state disclosure, vertical and temporal resolution, and matched observations. Effective information emerges when numerical diagnostics, archive design, and observed support are evaluated together. This criterion converts resolution from a nominal property into a testable link in validation architecture. Those distinctions require a validation architecture that records how each modeled scale is mapped to each observed object.

6 Validation architecture for WRF macroburst studies

Validation begins by defining the comparison target before model performance is examined. A Childs gust at a stated height and cadence needs a declared mapping to WRF 10 m wind, lowest-level instantaneous wind, or a diagnostic gust. Radar reflectivity needs a forward operator or bounded surrogate because model hydrometeor content is a different quantity. Damage direction constrains spatial organization but does not resolve velocity, and a model sounding supplies modeled context rather than an observation. These mappings define the quantities to be evaluated and prevent an apparent agreement from joining incompatible variables.

Environmental reconstruction establishes whether a candidate event can support the process being tested. For Akron, stable postfrontal stratification, low-level shear, moisture, inversion, and critical-layer structure belong to this layer. Romanic et al. (2022) provide report-based environmental classes and hourly ERA5 context for United States downbursts, while Hadavi and Romanic (2024a) derive a Canadian precursor parameter from reanalysis and report classes. These sources can stratify sensitivity tests, but their spatial and temporal support differs from the realized Akron profile, and treating their discrimination metrics as event truth would erase precisely the environmental uncertainty that the WRF experiment needs to evaluate.

Storm initiation, organization, path, and timing determine whether the simulation contains the target event object. Radar and surface networks establish this gate. In the Ligurian episode, the modeled derecho and the missed earlier cell coexisted, so agreement with the later system could not validate the local storm (Burlando et al., 2024, pp. 6 to 8). Akron comparison likewise needs the elevated system to reach the correct region and time with relevant banding. Reporting displacement and timing separately preserves the difference between a structurally useful translated analysis and a forecast-like event reconstruction.

Once event identity is established, the surface sequence can be evaluated in a common time base. Wind magnitude, duration, direction, pressure, temperature, and precipitation jointly constrain the reconstruction. The corrected Childs sequence, including the 1.5 °C rise concurrent with the maximum gusts and the subsequent 3.5 °C fall, is more restrictive than any isolated pressure or wind maximum. Borchardt et al. (2024) found related pressure oscillations and directional behavior in other elevated-convection cases, while similar radar signatures also occurred without documented damage. Component-level results therefore preserve partial agreement and alternative interpretations better than one deterministic signature.

Footprint comparison adds spatial support and measurement geometry. Hjelmfelt (1988)

observed pronounced asymmetry and distinct lifetimes for individual and line-type microbursts. Fujita (1985) showed how radar blockage, station placement, and damage evidence can alter inferred maxima. In mechanically forced laboratory flows, Romanic and Hangan (2020) found nonlinear interaction between a downburst-like outflow and background boundary-layer flow, but source-momentum differences and reduced-scale conditions constrain that comparison. Together these sources justify testing asymmetry, translation, and sampling without functioning as direct WRF validation.

Vertical structure and mechanism constitute the most demanding scientific layer. Relevant Akron diagnostics include wave trapping, vertical energy propagation, critical-layer behavior, overturning, pressure-gradient acceleration, parcel origin, negative buoyancy, hydrometeor cooling, and momentum transport. Schumacher et al. (2023) supply wave and trajectory diagnostics, Li et al. (2023) supply event-specific pressure-work and evaporative-cooling analysis, and Liu et al. (2023) supply momentum-tendency and hydrometeor-process analysis. Joint evaluation asks whether a diagnosed chain is consistent with the event environment and surface sequence without searching for one mechanism that must govern every destructive wind.

Comparator uncertainty remains part of each result. Wakimoto (1982) used radar and rawinsondes to distinguish gust-front stages and pressure components, but two-dimensional retrieval and sampling-height assumptions constrain mapping to a three-dimensional field. He et al. (2024) combined a profiler and near-ground anemometer for four Hong Kong events, revealing nonstationary profiles and elevated maxima while retaining separation between instruments and complex-site effects. Borchardt et al. (2024) documented report and radar-geometry limitations in its catalog. These uncertainties divide mismatch among model, observation, and mapping instead of assigning it automatically to WRF.

Independent audit also depends on reproducibility. The numerical record identifies the WRF version and executable, source and lateral forcing, initialization, projection, domain extent, nesting, and horizontal and vertical coordinates. It separately records time steps, physics and call intervals, assimilation, damping and filtering, output cadence, postprocessing, and observation operators. Skamarock et al. (2019) define the technical meaning of many fields but offer no event-specific package. When a publication omits part of the record, the omission becomes a documented uncertainty and is not filled by assumed defaults.

Figure 4 summarizes the resulting sequence. Partial, failed, and not-reported states are retained, and the resulting transfer product is a candidate boundary field, not an approved inlet or design quantity.

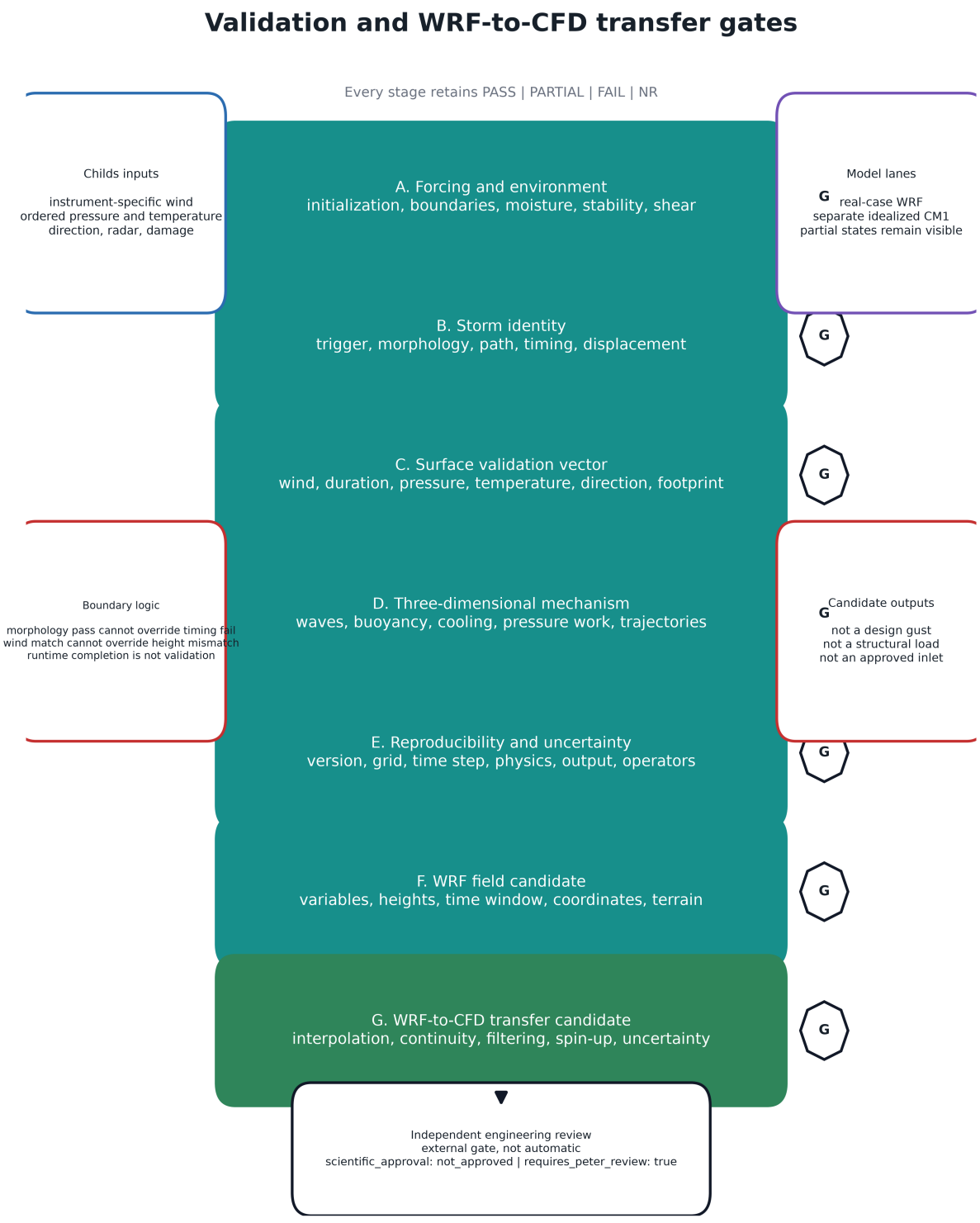


Figure 4: Validation and transfer gates. Sequential stages connect environment and forcing, storm identity, multivariable surface sequence, footprint, vertical mechanism, uncertainty and reproducibility, and a candidate boundary-plane extraction. Failed or partial gates remain visible. Engineering use occurs only after height, averaging, filtering, coordinates, continuity, and independent review are satisfied. The framework remains proposed and not approved.

This architecture is a proposed research framework, not an approved standard. Its logic is cumulative: environmental and morphology agreement define a plausible event target; surface-sequence agreement narrows local error; vertical diagnostics test competing accelerations; reproducibility allows independent audit; and a separate transfer review evaluates downstream admissibility. Mechanism discrimination is the most demanding validation layer because several processes can yield similar surface divergence while requiring different environmental and vertical evidence.

7 Competing and complementary mechanisms

Four mechanism families recur across the corpus, and their surface expressions can overlap. The first centers on negative buoyancy from evaporation, melting, and hydrometeor loading followed by impingement and radial spreading. Proctor (1988) found rain evaporation dominant and hail melting secondary in isolated axisymmetric TASS simulations, together with ring-vortex evolution and frictional shaping. Proctor (1989) showed sensitivity to loading, shaft size, hydrometeor type, duration, environmental stability, and imposed rotation. These studies establish process hypotheses under TASS equations and forcing; their grids, imposed precipitation, and axisymmetry do not validate WRF or describe Akron's elevated parent storm.

Cold-pool and gust-front dynamics form a second family. Wakimoto (1982) related pressure structure and propagation to life-cycle stage, separating postfrontal hydrostatic pressure from a prefrontal stagnation-pressure feature. Rotunno et al. (1988) used an idealized predecessor cloud model to connect squall-line maintenance with the balance between cold-pool circulation and low-level shear. Weisman (1992) examined rear-inflow-jet formation and its effects on long-lived mesoscale convective systems. Johns and Hirt (1987) and Ashley and Mote (2005) characterize the broad derecho hazard object. These results explain storm organization and momentum pathways at their respective observational or idealized scales, becoming candidate diagnostics for WRF output rather than WRF skill evidence.

The Akron evidence motivates a third, gravity-wave-centered family. Childs et al. (2021, pp. 4 to 7) documented an event with elevated convection above stable near-surface air. The same event record contains a short temperature rise before the gust and pressure surge, subsequent cooling, rapid pressure evolution, and outflow evidence (Childs et al., 2021, pp. 1, 13). These constraints leave a minimal cold-pool account incomplete; independent evidence is still required to establish wave breaking. Schumacher et al. (2023) add simulated trapping between a stable layer and a critical layer, amplification, nonlinear breaking, trajectories, and repeated downbursts. Agreement across the observation and model branches makes

the mechanism physically plausible for Akron, while pressure-phasing error, configuration sensitivity, and overstrong free-slip CM1 winds bound the interpretation.

Borchardt et al. (2024) extend wave-coincident damaging-wind evidence beyond Akron but preserve a key conflict. Its WRF case showed downward wave branches and surface responses with no clear breaking. The evidence supports a spectrum in which wave motion can contribute to damaging wind and breaking is a candidate for more extreme outcomes. The Childs pressure-temperature sequence, stable environment, surface wind, and vertical diagnostics are jointly necessary for testing the more specific Akron chain.

Near-ground structure and cross-model transfer define a fourth family of modifiers, not a meteorological cause by themselves. Qu and Ji (2013) documented irregular, nonaxisymmetric, multicenter fields in an ARPS-to-cloud-model-to-FLUENT workflow. Romanic and Hangan (2020) demonstrated nonlinear interaction between a mechanically forced downburst-like outflow and ambient boundary-layer flow under reduced-scale laboratory conditions. These sources support multicell structure, ambient-flow interaction, and transfer choices as sources of asymmetry within their respective systems but do not isolate a single demonstrated cause of the reported asymmetry.

Mechanism synthesis is consequently conditional. Evaporative or melting-induced negative buoyancy, trapped-wave amplification and breaking, cold-pool and gust-front circulation, pressure-gradient acceleration, vertical momentum transport, and rear inflow can act in different proportions and can coexist. A credible diagnosis links event-specific environment to three-dimensional tendencies and then to the synchronized surface sequence. The Childs vector provides those surface and environmental constraints, Schumacher et al. (2023) provide one bounded diagnostic path, and the wider corpus supplies alternatives that can challenge it. Once a mechanism is judged scientifically plausible at event scale, a separate problem begins: determining whether the reconstructed field carries the information needed for cross-model and engineering transfer.

8 From WRF fields to CFD and engineering use

A WRF field is a candidate mesoscale description, not automatically a CFD inlet. The transfer crosses model equations, grid scale, terrain representation, turbulence closure, coordinates, temporal filtering, and boundary conditions. A valid interface must first preserve event identity: if the model system is displaced or delayed, extracting a boundary plane at the observed site and time can combine the wrong storm object with the correct coordinates. Translating the simulated event to align with observations may be analytically useful, but the adjustment must be declared and its implications for terrain, forcing, and uncertainty

retained.

Height and variable identity are immediate gates. WRF 10 m wind, lowest-level wind, diagnostic gust, and instantaneous resolved wind are not interchangeable. The Childs maxima belong to multiple heights and instruments. A CFD inflow plane needs vertical profiles and evolving direction, not one peak. Pressure, temperature, density, moisture, and turbulence variables may also be needed depending on the downstream solver and objective. Childs et al. (2021) constrain selected near-surface series but do not observe the complete plane. Schumacher et al. (2023) supply model fields but not their independent engineering validation.

Temporal mapping preserves nonstationarity by recording the model integration step, archive cadence, any gust diagnostic window, the CFD time step, and the averaging applied at each interface. The Akron pressure and wind changes occur on seconds-to-minutes scales, while many WRF case archives use minute-scale or coarser output. Interpolation cannot reconstruct unresolved dynamics. Directional rotation and translation can dominate the load history even if the maximum speed is similar.

Spatial filtering and coordinates require equal care. A WRF grid value represents a numerically filtered volume influenced by map projection and terrain. A CFD inlet may be a vertical plane in local Cartesian coordinates with much finer cells. Conservative interpolation, vertical datum, surface elevation, roughness transition, and topographic consistency must be defined. Romanic and Hangan (2020) indicate that background-flow interaction can be nonlinear, while He et al. (2024) show that nonstationary wind structure can differ from conventional stationary spectra. These sources motivate interface tests rather than supplying direct inlet prescriptions.

Field continuity and balance are also scientific requirements. A snapshot assembled from independently interpolated variables may violate mass continuity, thermodynamic consistency, or pressure-velocity relationships. Time-dependent boundaries can generate numerical transients in CFD. Qu and Ji (2013) provide a one-way cross-model analogue and record differences between cloud-model and CFD winds. Because its upstream model is ARPS and its chain omits two-way feedback, it illustrates transfer risk rather than validating a WRF-to-CFD method.

Engineering outputs demand a further evidence layer. A simulated peak wind does not become a design gust without a defined averaging interval, exposure, height, directional convention, event probability, and approved statistical framework. The Zhejiang and Genoa sources involve infrastructure consequences, but Li et al. (2023) do not provide tower fragility or a measured tower-site wind history, and Burlando et al. (2020) lack colocated bridge measurements and causal proof. Their WRF fields cannot be promoted to structural validation.

Climatology and machine learning do not close that gap. Romanic et al. (2022) and Hadavi and Romanic (2024a) describe environments and report-conditioned discrimination. Hadavi and Romanic (2024b) trained classical machine-learning classifiers on ERA5-derived variables for Canadian downburst-prone environments, with substantial class imbalance and reporting bias. Forecasting indices tested by Bolgiani et al. (2020b, p. 10) produced AUC values near 0.49 to 0.51, providing no useful microburst discrimination in the tested set. Trapp et al. (2010, pp. 8 to 9) evaluated a broad severe-convection proxy at aggregated scales. These studies can support event selection or uncertainty stratification without supplying local gust return periods, event-resolved three-dimensional fields, or calibrated engineering boundaries.

A WRF-to-CFD candidate therefore carries a provenance manifest identifying the event and WRF run, model and executable identity, extraction time and location, variables and heights, and original and transferred grids. A linked interface record then identifies temporal and spatial filters, coordinate transforms, terrain and roughness treatment, and continuity checks. Uncertainty fields, validation status against each Childs-vector component, and explicit prohibited uses complete the record. Independent scientific and engineering review governs any use beyond this documented candidate state.

9 Research agenda centered on the Childs event

The Akron program first recovers the source configuration before any optimization against the observations. WRF version, initial and lateral data, domain geometry, vertical levels, time steps, physics, nesting, output, and postprocessing reported by Schumacher et al. (2023) are recorded with file hashes and execution identity. Fields absent from the source packet are recorded as not reported (NR). This reproducibility stage determines whether the numerical workflow can be repeated and whether its outputs are complete; event-level scientific evaluation begins only after those outputs are mapped to the Childs vector.

Observation operators define that mapping before sensitivity results are inspected. Each instrument record carries location, height, sampling response, time standard, averaging, quality control, and representativeness. Radar comparison carries beam geometry or a bounded surrogate, while damage conveys spatial and directional support but not instantaneous speed. Environmental analysis distinguishes direct measurements from RAP, WRF, or other modeled profiles. These declarations prevent a favorable lowest-level wind or shifted grid maximum from silently substituting for the observed 10 m and 2.3 m tower records.

The baseline comparison preserves partial and failed dimensions. Event timing and displacement, storm morphology, the corrected pressure-temperature sequence, wind magnitude

and duration, direction, footprint, and vertical mechanism are evaluated separately. The Schumacher pressure-phasing discrepancy is an explicit target for reproduction or explanation (Schumacher et al., 2023, p. 8, Fig. 8). When spatial or temporal translation is useful for process analysis, raw and adjusted results retain separate status because structural similarity after alignment is not forecast-like timing or location skill.

Sensitivity work then follows identifiable uncertainty pathways. Grid and vertical resolution affect the wave duct and diagnosed overturning. PBL treatment and lower-boundary drag affect stable-layer preservation, shear, mixing, and surface stress. Microphysics changes heating, cooling, and hydrometeor tendencies. Forcing, initialization, lateral-boundary placement, damping, and archive cadence affect trigger placement and what can be compared. Cohen et al. (2015), Bolgiani et al. (2020a), and Bolgiani et al. (2020b) justify these axes without selecting a preferred Akron branch. Evaluation against the full Childs vector prevents a larger peak wind from compensating invisibly for worse timing, phasing, or footprint.

Mechanism testing is specified at the diagnostic level. Wave trapping requires an environmental structure that restricts energy propagation. Amplification and breaking require amplitude, overturning, and trajectory evidence beyond banded morphology. Pressure-gradient acceleration, buoyancy, hydrometeor tendencies, and momentum transport use common time windows that connect them to the station sequence. Cooling and cold-pool alternatives are retained until diagnostics exclude them. WRF and CM1 analyses continue as separate branches; CM1 isolates sensitivity to the lower boundary, environmental profile, and idealized trigger and is not treated as a higher-resolution WRF surrogate.

Cross-event analysis tests recurrence while preserving heterogeneity. The elevated-convection family documented by Borchardt et al. (2024) supplies cases with different radar geometry, surface sampling, and damage reporting. Direct WRF cases outside Akron test whether the same validation architecture remains useful under cooling-dominated or organized-system regimes. The objective is recurrence of diagnostic combinations under declared conditions, not a universal macroburst label.

Effective-resolution analysis accompanies nominal refinement through spectra, filtering, convergence of relevant diagnostics, and sensitivity to archive cadence. Bolgiani et al. (2020b) show that fine domains can become overenergetic within a specific configuration. Akron experiments need to establish whether the stable wave duct, pressure sequence, and near-surface response converge in physically interpretable ways. TASS and idealized cloud-model benchmarks can motivate diagnostics only while their equations, forcing, and closures remain separate.

An engineering branch opens after an event-level mesoscale field satisfies defined scientific criteria. Its first artifact is a provenance-rich WRF field packet, not a design wind. Offline

extraction and filtering can then be tested for conservation, nonstationarity, coordinate and terrain treatment, and uncertainty. CFD sensitivity to plane location, vertical mapping, and turbulence treatment can be studied while keeping the incoming mesoscale field status unchanged. Structural and exposure evidence, followed by independent engineering review, controls any later design interpretation.

The agenda is cumulative because reproduction, observation mapping, event validation, mechanism testing, cross-event analysis, effective-resolution assessment, and transfer answer different questions. Current Schumacher reproduction consists of planning and workflow-readiness artifacts; scientific evidence requires source-locked execution and an independent comparison with the Childs vector. Scientific status is `not_approved`, and configuration, mechanism, transfer, and engineering interpretations require PI review. These staged tasks follow from four broader tensions that the Discussion now integrates across the corpus rather than repeating the case sequence.

10 Discussion

The first cross-cutting tension is reconstruction versus local truth. WRF can join synoptic forcing, mesoscale boundaries, storm organization, and three-dimensional diagnostics in ways that point observations cannot. That integrative strength is conditional on initialization, boundaries, physics, numerics, and event placement. The corpus repeatedly shows that a useful parent-system reconstruction can coexist with a wrong local event. The Mediterranean hindcast captured the derecho but missed an earlier damaging cell (Burlando et al., 2024, pp. 6 to 8); the Genoa branches improved selected winds while retaining delay and noncolocated observations (Burlando et al., 2020, pp. 23 to 24, 28 to 29); the Shanghai case retained footprint and decay-stage errors (Liu et al., 2023, pp. 6 to 8); and the Zhejiang case retained an intensity error alongside informative diagnostics (Li et al., 2023, pp. 3 to 4). Childs et al. (2021, p. 1) expose the consequence: local truth is not one maximum but a synchronized, instrument-specific sequence. A correct regional environment with a displaced trigger is evidence about environmental plausibility, not local validation. A local maximum after translation can support process analysis but not timing or location skill. The useful unit of evidence is a pattern of passes, partial matches, failures, and unresolved mappings, not a scalar success label.

The second tension is nominal refinement versus effective information. Akron required the 267 m WRF nest to expose severe local structure absent from the 4 km domain, while the MIST spectral experiment found overenergetic behavior in 1.2 km and 400 m domains and placed the 3.6 km domain's effective resolution near 30 km for that configuration and

diagnostic (Bolgiani et al., 2020b, pp. 8, 11). These outcomes are compatible because cell size is only one component of the information pathway. Vertical discretization, time integration, diffusion, physics coupling, archive cadence, observation support, and event placement determine whether added structure is interpretable. Historical TASS and CM1 grids cannot supply a WRF threshold because their equations, forcing, and boundary assumptions differ. Refinement should therefore be judged by whether relevant diagnostic quantities become more stable and comparable: for Akron, that means convergence of the wave duct, pressure-temperature phasing, wind footprint, and observation-operator outputs instead of visual detail alone.

The third tension is convergent mechanism evidence versus preserved model identity. The Childs-Schumacher chain is strong because observations, real-case WRF, and idealized CM1 address different portions of one hypothesis. Childs et al. (2021) constrain the event sequence, WRF reconstructs wave-related processes under real-event forcing, and CM1 isolates their behavior under controlled assumptions. Agreement in environmental structure, wave-like response, pressure organization, and intense surface wind increases plausibility, while the WRF pressure-phasing mismatch and excessive free-slip CM1 winds remain falsifiable conflicts. Convergence adds information because the contributing evidence sources are kept distinct. The wider corpus prevents that bounded convergence from becoming a universal mechanism: TASS emphasizes evaporation, melting, and ring-vortex dynamics under imposed isolated forcing; gust-front and predecessor cloud-model studies emphasize cold-pool, shear, and rear-inflow organization; direct Shanghai and Zhejiang WRF analyses emphasize hydrometeor, pressure-gradient, advection, and momentum pathways; and Borchardt et al. (2024) find wave-coincident damaging winds without clear breaking in their modeled case.

The fourth tension is scientific reconstruction versus engineering admissibility. Mesoscale validation often stops at radar morphology, station wind, or a mechanism cross section, whereas a CFD boundary requires coherent, time-varying fields at declared heights and coordinates. Childs et al. (2021) supply exceptional surface constraints without a complete plane. Schumacher et al. (2023) supply model fields without independent engineering validation. Qu and Ji (2013) and Romanic and Hangan (2020) show, under different model and laboratory identities, that cross-model transformation, multicell structure, and ambient-flow interaction can alter near-ground flow. Climatology and machine learning enter this chain upstream rather than closing it: Romanic et al. (2022), Hadavi and Romanic (2024a), Hadavi and Romanic (2024b), Johns and Hirt (1987), and Ashley and Mote (2005) can classify environments, parent systems, occurrence contexts, or consequences, while report bias, reanalysis support, class definition, imbalance, and aggregation prevent those products from validating an Akron gust or WRF mechanism. Their appropriate contribution is stratification rather

than validation.

Across all four tensions, a better simulation preserves more of the event identity, multivariable sequence, footprint, and physically diagnosed chain while reporting how initialization, mapping, and uncertainty condition that result. A better transfer records what its arrays represent, which comparisons succeeded, which scales are unresolved, and which uses require separate review. This framing supports substantial scientific use of WRF while keeping forecast authority, CFD candidacy, and engineering approval as distinct evidentiary states.

11 Concluding remarks

Childs et al. (2021) transform WRF macroburst evaluation from a peak-wind comparison into a coordinated test of event identity, instrument height, cadence, pressure-temperature phasing, directional evolution, radar morphology, damage footprint, and environmental structure. The Childs sequence records a 1.5 °C rise concurrent with the maximum gusts followed by a 3.5 °C fall, while the 50.45 and 51.12 m s^{-1} maxima are tied to the 10 m and 2.3 m towers (Childs et al., 2021, p. 1). Schumacher et al. (2023) supply a compelling but bounded bridge from those observations to trapped gravity-wave amplification and breaking through real-case WRF and separate idealized CM1 branches. Agreement in wave-like structure, stable shear, pressure organization, and intense surface wind strengthens plausibility, while the station-pressure mismatch, configuration dependence, single-event scope, and overstrong free-slip CM1 winds materially constrain the interpretation. Other WRF cases show that parent-system reconstruction can coexist with errors in local trigger, timing, footprint, intensity, or decay, and the resolution evidence shows that nominal refinement cannot replace effective-resolution, cadence, physics, and observation-operator tests.

The defensible route from mesoscale reconstruction to engineering use is an evidence chain with explicit scientific criteria. Environmental and storm identity precede local multivariable comparison; local comparison precedes mechanism attribution; mechanism attribution precedes boundary-field candidacy; and a provenance-rich WRF field is distinct from a CFD inlet, design gust, or structural load. Historical TASS, ARPS, idealized cloud-model, observational, climatological, machine-learning, and wind-tunnel sources sharpen mechanisms and validation targets but do not directly validate WRF. The immediate Akron agenda is source-locked reproduction, predefined Childs observation operators, visible partial and failed comparisons, targeted resolution and physics sensitivity, separate WRF and CM1 diagnostics, and independent review before transfer. Current reproduction activity documents planning and workflow readiness, scientific status is `not_approved`, and PI review is required before any configuration, mechanism, transfer, or engineering interpretation is promoted.

Declarations

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Data availability. This review is based on the literature cited in the References. The PETW claim, citation, source, and review ledgers are project-local research records and are not presented as a public scientific dataset. Any external data-availability statement requires PI confirmation.

Author contributions. Author roles require PI confirmation before external submission.

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